



**CALICUT UNIVERSITY – FOUR-YEAR UNDER  
GRADUATE PROGRAMME (CU-FYUGP)**

**BSc CHEMISTRY**

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |                   |                    |             |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------|--------------------|-------------|
| Programme      | B. Sc. Chemistry                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |                   |                    |             |
| Course Title   | <b>INSTRUMENTAL METHODS OF ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                                                                                          |                  |                   |                    |             |
| Type of Course | <b>MAJOR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  |                   |                    |             |
| Semester       | <b>VII</b>                                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |                   |                    |             |
| Academic Level | <b>400-499</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |                   |                    |             |
| Course Details | Credit                                                                                                                                                                                                                                                                                                                                                                                                                                           | Lecture per week | Tutorial per week | Practical per week | Total Hours |
|                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                | -                 | 2                  | 75          |
| Pre-requisites | 1. Fundamentals of Electrochemistry.<br>2. Fundamentals of Spectroscopy.<br>3. Fundamentals of Analytical Chemistry.                                                                                                                                                                                                                                                                                                                             |                  |                   |                    |             |
| Course Summary | This course provides a thorough overview of essential analytical techniques in chemistry and materials science, covering separation, spectroscopy, surface characterization, and thermal/electroanalytical methods. The course emphasizes practical applications, preparing students for precise chemical analysis in various scientific and industrial fields. The practical module ensures hands-on-training on some of the important methods. |                  |                   |                    |             |

**Course Outcomes (CO):**

| CO  | CO Statement                                                                                                                               | Cognitive Level* | Knowledge Category# | Evaluation Tools used                       |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|---------------------------------------------|
| CO1 | To understand the basic principles and instrumentation of chromatographic techniques for the separation of mixture of chemicals (products) | Ap               | P                   | Instructor-created exams / Quiz /Assignment |
| CO2 | To identify the instrumentation and applications of important spectroscopic methods for chemical analysis.                                 | Ap               | P                   | Class test /Assignment /Quiz                |
| CO3 | To know the role of imaging techniques to study various materials and surfaces.                                                            | U                | C                   | Assignment/ Class test                      |
| CO4 | To understand and analyse the principles and instrumentation of various thermal analytical methods                                         | An               | C                   | Assignments /Seminar presentation           |

|                                                                                                                                                                                                   |                                                   |    |   |                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----|---|-----------------------------------|
| CO5                                                                                                                                                                                               | To understand and apply electroanalytical methods | Ap | C | Assignments /Seminar presentation |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)<br>Metacognitive Knowledge (M) |                                                   |    |   |                                   |

### Detailed Syllabus:

| Module    | Unit                                     | Content                                                                                                                                                                                                                                                                                               | Hrs       | Marks     |
|-----------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| <b>I</b>  | <b>Separation techniques</b>             |                                                                                                                                                                                                                                                                                                       | <b>6</b>  | <b>10</b> |
|           | 1                                        | Brief outline of Paper and Thin-Layer Chromatography (TLC),                                                                                                                                                                                                                                           | 1         |           |
|           | 2                                        | Ion exchange chromatography: Principle, cation and anion exchange resins, its application in separation of ions.                                                                                                                                                                                      | 1         |           |
|           | 3                                        | Gas Chromatography (GC) – Principle, GC Instrumentation – Injectors, column, detectors (TCD, FID, ECD) and applications.                                                                                                                                                                              | 1         |           |
|           | 4                                        | Hyphenated GC Technique - GC-MS                                                                                                                                                                                                                                                                       | 1         |           |
|           | 5                                        | High Performance Liquid Chromatography (HPLC): Principle, instrumentation - Column, stationary phases, column packing, mobile phase, detectors. Effects on Separation of Composition of the Mobile Phase and applications                                                                             | 2         |           |
| <b>II</b> | <b>Spectroscopic and related methods</b> |                                                                                                                                                                                                                                                                                                       | <b>15</b> | <b>36</b> |
|           | 6                                        | UV-Visible Spectrometry: Beer-Lambert's law, Basic principles of instrumentation (choice of source, monochromator and detector) for single and double beam instruments. Difference between Colorimeter & spectrophotometer.                                                                           | 2         |           |
|           | 7                                        | Flame emission and Atomic Absorption Spectroscopy (AAS): Introduction, principle. Instrumentation and Analytical applications. Inductively Coupled Plasma-Atomic/Optical emission spectroscopy (ICP-AES or ICP-OES)- theory, instrumentation and applications. ICP-MS method. Analytical Applications | 3         |           |
|           | 8                                        | Fourier transform-Infrared spectroscopy (FT-IR)- FT-IR instrumentation and analytical applications. Raman spectroscopy, Principle, instrumentation, Surface enhanced Raman Spectroscopy (SERS), Raman microscopy                                                                                      | 3         |           |

|            |                                                                                |                                                                                                                                                                                                                                                            |           |           |
|------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
|            | 9                                                                              | Fluorescence spectroscopy, Theory, Instrumentation and Analytical applications, Confocal laser-scanning microscopy                                                                                                                                         | 2         |           |
|            | 10                                                                             | FT-NMR spectroscopy, basic principle, instrumentation, spectrometers with different frequencies of operation, ESR/EPR spectroscopy, ENDOR (Electron-Nuclear double resonance) technique                                                                    | 3         |           |
|            | 11                                                                             | Mass spectrometry, Instrumentation and applications, MALDI-TOF method and instrumentation                                                                                                                                                                  | 2         |           |
| <b>III</b> | <b>Microscopy, Photoelectron spectroscopy and X-ray diffraction techniques</b> |                                                                                                                                                                                                                                                            | <b>14</b> | <b>32</b> |
|            | 12                                                                             | Scanning Electron Microscopy (SEM) Instrumentation, Operating Principle, Secondary - Electron Images, Backscattered- Electron Images, operating conditions and sample preparation.                                                                         | 2         |           |
|            | 13                                                                             | Transmission Electron Microscopy (TEM): Instrumentation, General Design, Resolution, Electron Sources, TEM grids, electron lenses, Bright and Dark field images, Applications.                                                                             | 2         |           |
|            | 14                                                                             | Scanning probe microscopy methods: Principle, instrumentation and applications of Scanning Tunneling Microscopy (STM) and Atomic Force Microscopy (AFM).                                                                                                   | 3         |           |
|            | 15                                                                             | Photoelectron Spectroscopy Techniques - X-Ray Photoelectron Spectroscopy - Instrumentation for XPS, Sample Introduction and Handling for Surface Analysis, Analytical Applications of XPS, Auger Electron Spectroscopy – Instrumentation and Applications. | 3         |           |
|            | 16                                                                             | Powder and Single crystal X-ray diffraction, basic principle, instrumentation and applications.                                                                                                                                                            | 4         |           |
| <b>IV</b>  | <b>Thermal and Electroanalytical methods of analysis (12 h)</b>                |                                                                                                                                                                                                                                                            | <b>10</b> | <b>20</b> |
|            | 17                                                                             | Thermogravimetry - TGA Instrumentation, Analytical Applications of Thermogravimetry, Derivative Thermogravimetry, Sources of Error in Thermogravimetry.                                                                                                    | 2         |           |
|            | 18                                                                             | Differential Thermal Analysis (DTA) - Instrumentation - Analytical Applications of DTA.                                                                                                                                                                    | 1         |           |
|            | 19                                                                             | Differential Scanning Calorimeter (DSC), Instrumentation and applications.                                                                                                                                                                                 | 1         |           |
|            | 20                                                                             | Classification of electroanalytical methods, Potentiometry- Three and Two electrode systems, Types of indicator Electrodes. Analytical Applications of Potentiometry.                                                                                      | 2         |           |
|            | 21                                                                             | Coulometry – Electrogravimetry, Instrumentation for Electrogravimetry and Coulometry, Applied Potential, Analytical Determinations Using Faraday's Law.                                                                                                    | 2         |           |

|          |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |  |
|----------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--|
|          | 22                | Cyclic Voltammetry, Theory Instrumentation and applications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2         |  |
| <b>V</b> | <b>Practicals</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>30</b> |  |
|          | <b>Open Ended</b> | <p>At least 5 practical experiments must be performed from the given below list.</p> <ol style="list-style-type: none"> <li>1. Evaluation of the refractive index of the given liquid and also find its molar refractivity</li> <li>2. Determination of the order of a reaction and velocity constant for the inversion of cane sugar by acid by polarimetric method</li> <li>3. Study of the complex formation between ferric ion and salicylic acid to find the formula and stability constant of the complex via colorimetry</li> <li>4. Preparation and characterization of silver/gold nanoparticles by uv-vis spectroscopy</li> <li>5. Estimation of band-gap for Cu nanoparticles using absorption spectroscopy</li> <li>6. Estimation of glucose via enzymatic method by using colorimetry</li> <li>7. Determination of Na and K ions in unknown solutions via flame photometric method</li> <li>8. Determination of calcium content of milk samples/unknown Calcium salt solutions using flame photometer</li> <li>9. Thermogravimetric analysis of a salt hydrate (such as <math>\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}</math>, <math>\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}</math>)</li> </ol> |           |  |

|  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |
|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  |  | <p>10. Powder X-ray diffraction measurement, indexing of patterns and determination of unit cell parameters of crystalline solids (like NaCl, KCl or any other)</p> <p>11. Synthesis of nanoparticles and estimation of crystallite size from powder X-ray diffraction patterns by Scherrer equation.</p> <p>12. Determination of the formal reduction potential (<math>E_0</math>) and n values for the <math>[\text{Fe}(\text{CN})_6]^{3-}/[\text{Fe}(\text{CN})_6]^{4-}</math> couple in 0.1M <math>\text{KNO}_3</math> from the 2mM cyclic voltammogram</p> <p>13. Determination of the concentration of unknown <math>\text{K}_3[\text{Fe}(\text{CN})_6]</math> solution using a calibration graph of concentration vs. peak height from cyclic voltammogram</p> <p>14. Separation of a mixture of amino acids by Thin Layer Chromatography (TLC) and identify the test amino acids by measuring their <math>R_f</math> values.</p> <p>15. Preparation of Silica nanoparticles by a one-sep process (Stöber process) and morphological analysis via scanning electron microscopy</p> <p>16. Open ended</p> <p>17. Open ended</p> <p>18. Open ended</p> |  |  |
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#### References:

1. D.A. Skoog, D.M. West, F.J. Holler, S.R. Crouch, Fundamentals of Analytical Chemistry, a. 9th Edn ., Cengage Learning., 2014.

2. D.A. Skoog, F.J. Holler, T.A. Nieman, Principles Of Instrumental Analysis, Engage Learning India Edn.
3. H. H. Willard, L. L. Merritt, Jr., J. A. Dean and F. A. Settle, Jr., Instrumental Methods of Analysis, 6th ed., CBS 1986.
4. Vogel's Text Book of Quantitative Organic Analysis, 2th ed. ELBS
5. Dr. B. K. Sharma, *Instrumental Methods of Chemical Analysis*, 3<sup>rd</sup> Edition 2004.
6. James W. Robinson, Eileen M. Skelly Frame, George M. Frame II, Undergraduate Instrumental Analysis, Seventh Edition, CRC Press.

**Mapping of COs with PSOs and POs :Correlation Levels:**

|      | PSO 1 | PSO 2 | PSO 3 | PSO 4 | PSO 5 | PSO 6 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|------|-------|-------|-------|-------|-------|-------|-----|-----|-----|-----|-----|-----|-----|
| CO 1 | 3     | 2     | -     | 3     | 3     | 2     | 3   | 2   | 2   | -   | 2   | -   | 1   |
| CO 2 | 3     | 2     | -     | 3     | 3     | 2     | 3   | 2   | 1   | -   | 2   | -   | 1   |
| CO 3 | 3     | 2     | -     | 2     | 3     | 2     | 3   | 2   | 1   | -   | 2   | -   | 1   |
| CO 4 | 3     | 2     | 1     | 2     | 3     | 3     | 3   | 2   | 1   | -   | 1   | -   | 1   |
| CO 5 | 3     | 3     | 2     | 2     | 3     | 3     | 3   | 2   | 1   | -   | 3   | -   | 1   |
| CO 6 | 3     | 3     | 1     | 3     | 3     | 3     | 3   | 2   | 1   | 2   | 3   | 2   | 3   |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| -     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

**Mapping of COs to Assessment Rubrics :**

|      | Internal Exam | Assignment<br>/viva | Practical skill<br>evaluation | End Semester Examinations |
|------|---------------|---------------------|-------------------------------|---------------------------|
| CO 1 |               | ✓                   | ✓                             | ✓                         |
| CO 2 |               | ✓                   | ✓                             | ✓                         |
| CO 3 |               | ✓                   | ✓                             | ✓                         |
| CO 4 |               | ✓                   | ✓                             | ✓                         |
| CO 5 |               | ✓                   | ✓                             | ✓                         |
| CO 6 |               | ✓                   | ✓                             | ✓                         |